

Retail Logistics & SLA Performance Portfolio Case Study

1. Project Overview

This portfolio case study analyzes an e-commerce retail logistics dataset to evaluate overall delivery performance, Service Level Agreement (SLA) compliance, and operational patterns across product weights and shipment methods.

As a junior analyst, I completed this project to apply practical Excel-based data cleaning, transformation, and analytical techniques to real-world logistics challenges. The objective of this document is to present the analysis cleanly, clearly, and professionally for stakeholders, hiring managers, and operational teams.

2. Business Objective

Prompt and reliable delivery is critical to customer satisfaction and operational efficiency in retail. The primary business objectives for this analysis are:

- **Evaluate SLA Performance:** Measure the overall on-time versus late delivery rates across 10,998 processed orders.
- **Identify Operational Risk Patterns:** Segment orders by weight and transit mode to spot where late deliveries concentrate.
- **Support Operational Improvement:** Highlight key findings and suggest targeted areas for logistics managers to investigate further.

3. Dataset Overview

The dataset contains **10,998 individual shipment records**. Each record includes operational details about the product, customer, order characteristics, and final delivery status.

Summary of Dataset Fields

Field Name	Description
ID	Unique identifier for each shipment record
Warehouse_block	Specific warehouse facility sector (A, B, C, D, F)

Mode_of_Shipment	Primary transit mode (Flight, Road, Ship)
Customer_care_calls	Total inquiries logged for the order
Customer_rating	Score assigned by the customer (1 to 5)
Cost_of_the_Product	Retail value of the ordered item
Prior_purchases	Count of previous orders made by the customer
Product_importance	Priority rating (Low, Medium, High)
Gender	Customer gender identifier
Discount_offered	Promotional price reduction applied to the order
Weight_in_gms	Net package weight in grams
Reached.on.Time_Y.N	Delivery SLA status (1 = On Time, 0 = Late)

4. Executive Summary

- **Overall SLA Baseline:** Out of 10,998 total orders, **59.67% (6,562 shipments)** arrived on time, while **40.33% (4,436 shipments)** missed their delivery SLA window.
- **Heavy Freight Concentration:** Over half of all orders (**54.25%**) fall into the Heavy weight category ($\$ > 4,000 \text{ text{g}\$}$).
- **Primary Concentration of Delays:** Heavy shipments account for **76.4% of all late deliveries** observed in the dataset.

- **Logistics Mode Share:** **Ship** is the dominant transit mode, carrying approximately **67.85%** of all total orders.
- **Key Risk Combination:** The combination of **Heavy weight items transported via Ship** accounts for **52.03% of all observed late shipments**.

5. Key Findings

1. **Substantial Delivery Breach Rate:** Roughly 4 out of every 10 orders fail to meet expected delivery SLAs, pointing to significant potential operational bottlenecks.
2. **Disproportionate Late Volume in Heavy Goods:** Heavy goods represent 54.25% of overall order volume but generate 76.4% of all recorded late deliveries.
3. **High SLA Compliance in Medium Weight Items:** Medium-weight items show an exceptionally high on-time delivery rate (99.9%), presenting an anomaly that requires data validation.
4. **Volume Concentration in Maritime Freight:** Ship transportation accounts for the majority of all shipments, making sea freight operations the largest driver of overall delivery performance metrics.

6. Weight Class & SLA Analysis

To examine how package weight relates to delivery timelines, the dataset was binned into three standard operational categories based on **Weight_in_gms**:

- **Light:** $< 2,000\text{g}$
- **Medium:** $2,000\text{g} - 4,000\text{g}$
- **Heavy:** $> 4,000\text{g}$

Delivery SLA Performance by Weight Class

Weight Class	Total Orders	Late Orders	On-Time Orders	Late %	On-Time %
Heavy	5,966	3,389	2,577	56.8%	43.2%

Light	3,245	1,046	2,199	32.2%	67.8%
Medium	1,787	1	1,786	0.1%	99.9%
Total	10,998	4,436	6,562	40.33%	59.67%

Key Analytical Takeaways

- **Heavy Weight Segment:** Heavy shipments show a significantly higher delay rate (**56.8%**) compared to the overall company baseline (**40.33%**). Heavy items account for **3,389 out of the 4,436 total late shipments (76.4%)**. Heavy shipments are strongly associated with observed delivery delays and represent a key area for operational review.
- **Light Weight Segment:** Light items perform better than average, achieving a **67.8% on-time rate**.
- **Medium Weight Segment:** Medium weight items show an **unusually high on-time rate of 99.9%** (1,786 out of 1,787 on time). This extreme pattern should be validated with data engineering or warehouse teams to confirm tracking accuracy before drawing definitive operational conclusions.

7. Shipment Mode Analysis

The dataset tracks three primary methods of transit: **Flight**, **Road**, and **Ship**.

Overall Shipment Volume by Mode:

- Ship : ~67.85% (Majority of total volume)
- Road : ~16.00%
- Flight : ~16.15%

Concentration of Late Deliveries

When isolating the **4,436 late shipments** across the network:

- **Heavy + Ship Orders:** This specific combination represents **approximately 52.03% of all late shipments** in the dataset.
- **Interpretation:** This figure indicates that more than half of all late orders occur within a single segment: heavy goods shipped by sea. It shows where late volume is concentrated; it does not mean an individual Heavy + Ship package has a 52.03% probability of being delayed.

8. Recommendations

Based on the patterns identified in the data, the following next steps are recommended for logistics stakeholders:

1. **Investigate Heavy Goods Handling:** Conduct an operational audit of loading, packaging, and carrier dispatch procedures for packages exceeding \$4,000\text{g}\$ to determine why delays concentrate in this category.
2. **Review Maritime Carrier SLAs:** Partner with ocean freight providers to review throughput timelines and port processing buffers, given that Heavy + Ship orders comprise over half of all delayed deliveries.
3. **Audit Data Quality for Medium Items:** Verify tracking logging practices for the Medium weight category (\$2,000\text{g} - 4,000\text{g}\$) to confirm that the 99.9% on-time metric reflects actual performance rather than a system recording bias.
4. **Establish High-Risk Order Monitoring:** Implement daily operational tracking for orders that combine high promotional discounts with heavy package weights to flag potential delivery bottlenecks early.

9. Limitations

To ensure sound business decision-making, the following analytical limitations must be considered:

- **Correlation vs. Causation:** The data shows that heavy shipments experience higher delay rates, but it **does not prove that package weight causes delays**. External factors—such as carrier handling constraints, customs processing times, or port congestion—were not captured in this dataset.
- **Data Anomaly:** The 99.9% on-time delivery result for Medium weight items is statistically unusual for real-world operations and warrants system validation.
- **Unmeasured Operational Variables:** Key logistics metrics—such as origin/destination distance, carrier names, seasonal weather disruption, and specific dispatch timestamps—were unavailable for analysis.

10. Tools & Skills

Software Used

- **Microsoft Excel:** Data cleaning, transformation, logical modeling, and reporting.

Practical Excel Functions & Features Applied

- **Data Hygiene & Cleaning:** Built structured Excel Tables, removed duplicate rows, applied standardized number formatting, and filtered edge cases.
- **Conditional Logical Formulas:** Applied `=IF()`, `=AND()`, and `=AVERAGEIFS()` formulas to calculate operational metrics and set up dynamic flag columns.
- **Dynamic SLA Risk Flag:** Designed a `Risk_Flag` column using `=IF(AND(Discount > Avg_OnTime_Discount, Reached_Late), "High Risk", "Normal")` to isolate high-value delayed orders.
- **PivotTables & Aggregation:** Summarized multi-variable data, calculated percentage shares of totals, and cross-tabulated weight classes against shipment modes.
- **Interactive Slicers:** Added dynamic visual filters for `Mode_of_Shipment` and `Customer_rating` to allow real-time scenario analysis.